

$$\nabla f = 0$$

Module 3

Chapter 3

Introduction to Gradient Descent

What does Gradient Descent do?

Gradient descent is an algorithm that numerically estimates where a function outputs its *lowest values*. This essentially means that it finds the *local minima* for the function, but *not* by setting ∇f to 0 as we have seen before. Instead, gradient descent *approximates* the solution. Furthermore, all it needs, in order to run, is a function's numerical output, no formula required.

This distinction about Gradient Descent is very important since this is what makes it very useful.

For example, if we had a simple function, say, $f(x) = x^2 - 4$, we could have easily set $\nabla f = 0$ and calculated that $x = 2$ *minimizes* $f(x)$. Or we could use *gradient descent* to get a numerical approximation, something like $x \approx 1.99999967$.

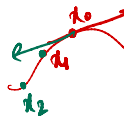
Both strategies arrive at the same answer.

But if we *don't* have a *simple formula* for our function, then it becomes hard or impossible to solve $\nabla f = 0$. If our function has a hundred or a million variables, then manipulating symbols isn't feasible. That's when an approximate solution is valuable, and gradient descent can give us these estimates no matter how elaborate our function is.

How does Gradient Descent work?

Think of a function $f(x, y)$ that defines some hilly terrain when graphed as a height map. Finding the local Maximum is computationally expensive owing to the complexity of the function. What can we do then?

We learned that the gradient evaluated at any point represents the direction of the steepest ascent up this hilly terrain. That might spark an idea for how we



$$x_1 = x_0 - \alpha \nabla f(x_0)$$

$$x_2 = x_1 - \alpha \nabla f(x_1)$$

could **maximize** the function: start at a random input, and as many times as we can, take a small step in the **direction of the gradient** to move uphill. In other words, walk up the hill.

$$x_2 = x_1 - \alpha \nabla f(x_1)$$

To **minimize** the function, we can instead follow the **negative of the gradient**, and thus go in the direction of steepest descent. This is **gradient descent**. Formally, if we **start** at a point x_0 and move a positive distance α in the direction of the negative gradient, then our new and improved x_1 will look like this:

$$x_1 = x_0 - \alpha \nabla f(x_0)$$

If we keep on taking our steps, then we can generalize:

$$x_{n+1} = x_n - \alpha \nabla f(x_n)$$

Here, α is also called the step size/learning rate. α plays a very important role in determining the training time of the algorithm as well as its precision.

How many iterations does gradient descent run for?

Normally gradient descent is run till the value of inputs does not change or the change in the value of inputs is below a certain threshold. The stopping criterion can also be a user defined maximum number of iterations.

Let's take an example:

(ref: <https://medium.com/@theom/a-very-short-visual-introduction-to-gradient-descent-with-code-204514d2153d>)

Let $f(x) = x^2$

We are standing at point $x_0 = -7.5$

Let's see how the gradient descent performs for three different learning rates:

Let $\alpha = 0.01, 0.1$ and 0.9

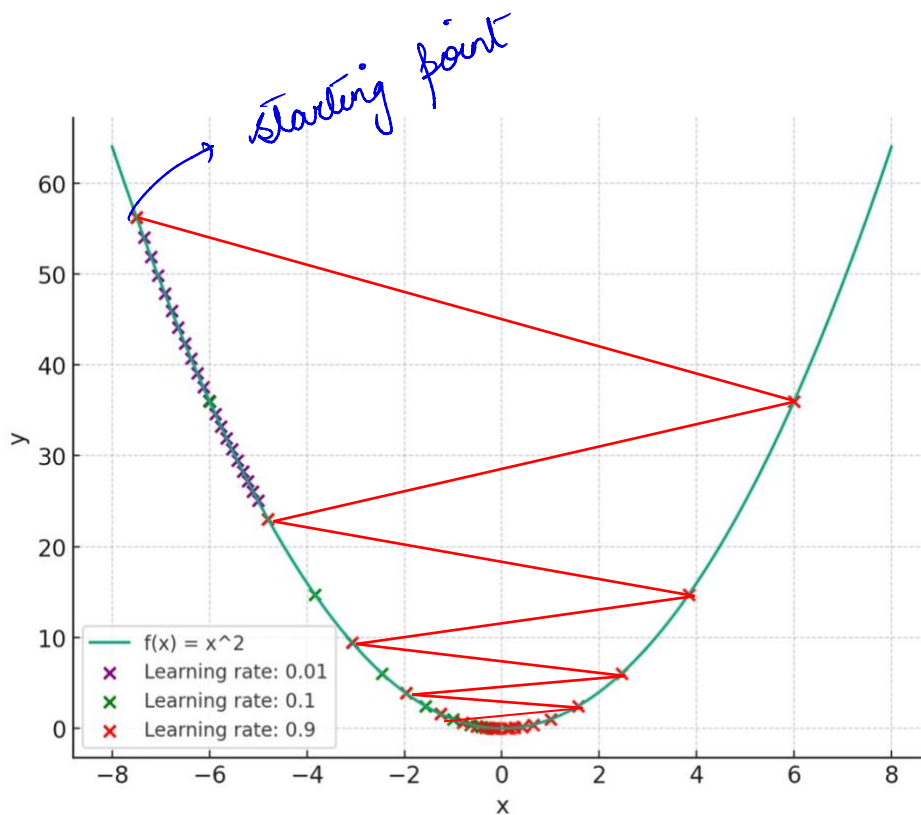


Fig 1

In Fig1, the plot shows color coded steps taken by the gradient descent algorithm for different learning rates. Let's dive into it:

- When the learning rate is too low (**0.01, purple crosses**), the algorithm makes very small steps and needs more time (or epochs) to reach the minimum.
- A moderate learning rate (**0.1, green crosses**) allows the algorithm to converge quickly to the minimum.
- A high learning rate (**0.9, red dots**) can cause the algorithm to overshoot the minimum and it becomes unstable.

This demonstrates the importance of the learning rate in the Gradient Descent algorithm. It's often a good idea to experiment with different learning rates to find

the one that allows the algorithm to converge in a reasonable amount of time without overshooting the minimum.

The learning rate can be fixed or variable. One of the most popular methods at the moment, is called *Adam*, which has a learning rate that scales over time.

Let's Practice

1. You have the following function:

$$f(x, y) = x^2 + 2y^2$$

You decided a learning rate of 0.1

Your initial starting point is (4,3)

Perform 2 steps of gradient descent and check if the value of your function is indeed reducing at each step.

2. For each function, find the gradient at (2,3), then perform one step of gradient descent, reporting the new values of x and y assuming a learning rate of 0.1. You can round answers to two decimal places.

- i. $f(x,y)=4x^2+3xy+2y^2$

- ii. $f(x,y)=4x/y$

3. If we were to use hill-climbing to minimize an error function $f(x,y)$, we might evaluate the function at some nearby points to the current guess (x_0, y_0) , then choose the smallest value, and repeat. Of course, if a function is differentiable, then we could do something faster: find the gradient and adjust in the opposite direction.

A kind of hybrid approach might be to find directional derivatives in the directions: $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$, $\begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$

This effectively samples eight equally spaced possible directions to adjust the location $\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ since the directional derivative just changes sign in the opposite direction. (This is definitely not better than gradient descent but we are just curious)

For each function $f(x,y)$, at the given location (x_0, y_0) :

1. Find these four directional derivatives (and negate them to get the directional derivative in the opposite direction), and note which of the 8 directions has the most negative slope.
2. Find the gradient of the function, and its negative.
3. Find the difference in f between the point that is 1 unit (learning rate) away in the best of the 8 directions (the one with the most negative slope), and the point that is 1 unit away in the direction of the negative gradient. If the two directions are identical, you can report 0 without evaluating f at the location.

i. $f(x, y) = xy^2 + 100$ at $(-1, 2)$

ii. $f(x, y) = x^3y^3 - x^2y + 1$ at $(1, 5)$

$$1. \quad f(x, y) = x^2 + 2y^2 \quad \underline{f(x_0, y_0) = 16 + 18 = 34}$$

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 4y \end{bmatrix}$$

$$\therefore \nabla f(4, 3) = \begin{bmatrix} 2 \times 4 \\ 4 \times 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 12 \end{bmatrix}$$

$$x_1 = x_0 - \alpha \nabla f(x_0) = 4 - 0.1 \times 8 = 4 - 0.8 = 3.2$$

$$y_1 = y_0 - \alpha \nabla f(y_0) = 3 - 0.1 \times 12 = 3 - 1.2 = 1.8$$

$$\therefore (x_1, y_1) = (3.2, 1.8)$$

$$\nabla f(3.2, 1.8) = \begin{bmatrix} 2 \times 3.2 \\ 4 \times 1.8 \end{bmatrix} = \begin{bmatrix} 6.4 \\ 7.2 \end{bmatrix}$$

$$x_2 = x_1 - \alpha \nabla f(x_1) = 3.2 - 0.1 \times 6.4 = 2.56$$

$$y_2 = y_1 - \alpha \nabla f(y_1) = 1.8 - 0.1 \times 7.2 = 1.08$$

$$(x_2, y_2) = (2.56, 1.08)$$

$$f(x_1, y_1) = (3.2)^2 + 2(1.8)^2 = 10.24 + 6.48 = 16.72$$

$$f(x_2, y_2) = (2.56)^2 + 2(1.08)^2 = 6.55 + 2.33 = 8.88$$

$\therefore$ The function is decreasing!

$$2) \rightarrow i) f(x, y) = 4x^2 + 3xy + 2y^2$$

$$\nabla f = \begin{bmatrix} 8x + 3y \\ 3x + 4y \end{bmatrix} \quad \therefore \nabla f(2, 3) = \begin{bmatrix} 25 \\ 18 \end{bmatrix}$$

$$\therefore x_1 = 2 - 0.1 \times 25 = 2 - 2.5 = -0.5$$

$$y_1 = 3 - 0.1 \times 18 = 3 - 1.8 = 1.2$$

$$- ii) f(x, y) = 4x/y$$

$$\nabla f = \begin{bmatrix} 4/y \\ -4x/y^2 \end{bmatrix} \quad \therefore \nabla f(2, 3) = \begin{bmatrix} 4/3 \\ -8/9 \end{bmatrix}$$

$$x_1 = 2 - 4/3 \times 0.1 = 1.87$$

$$y_1 = 3 - 0.1 \times (-8/9) = 3.09$$

$$3) \rightarrow f = xy^2 + 100 \text{ at } (-1, 2) \therefore \nabla f = \begin{bmatrix} y^2 \\ 2xy \end{bmatrix} \therefore \nabla f(-1, 2) = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \therefore \nabla_{\vec{u}_1} f = 4 \Rightarrow -\nabla_{\vec{u}_1} f = -4$$

$$\vec{u}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \therefore \nabla_{\vec{u}_2} f = -4 \Rightarrow -\nabla_{\vec{u}_2} f = 4$$

$$\vec{u}_3 = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} \therefore \nabla_{\vec{u}_3} f = 0 \Rightarrow -\nabla_{\vec{u}_3} f = 0$$

$$\vec{u}_4 = \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} \therefore \nabla_{\vec{u}_4} f = -4\sqrt{2} \Rightarrow -\nabla_{\vec{u}_4} f = 4\sqrt{2}$$

i) $\therefore$ Most negative out of these $\rightarrow \begin{bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} = \sqrt{2}/2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$$\text{ii) } \nabla f(-1, 2) = \begin{bmatrix} 4 \\ -4 \end{bmatrix} \therefore -\nabla f(-1, 2) = \begin{bmatrix} -4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

iii) We can see that $-\nabla f(-1, 2)$ is in the same direction as $\nabla_{\vec{u}_4} f(-1, 2)$ $\therefore$ The difference between two f values in the new location will be 0

